D.Y. PATIL COLLEGE OF ENGINEERING & TECHNOLOGY, KOLHAPUR

(An Autonomous Institute)

DEPARTMENT OF CSE (DATA SCIENCE)

A

Synopsis Report

on

AgriFlow AI — Smart Agricultural Supply Chain SaaS

Submitted by

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ABSTRACT

In most agricultural supply chains, farmers, buyers and transporters end up working separately from one another. A farmer may harvest a crop with no confirmed demand, while a buyer has a hard time finding produce that matches the crop type, quantity, quality and delivery date they need. When the crop is perishable, these delays quickly turn into spoilage, wastage and financial loss. AgriFlow AI is proposed as a smart agricultural supply-chain SaaS platform that brings all these people together in one digital workflow.

Farmers can list their crops along with the quantity, expected harvest date, location, price and quality. Buyers can search for produce, pre-book crops before they are harvested, or buy stock that is already harvested. An AI-based matching module then ranks suitable farmers for each buyer using price, quality, quantity, distance, harvest date, delivery needs and reliability.

To help with planning, a demand forecasting module studies past orders and seasonal patterns. For perishable crops, a shelf-life and wastage prevention layer gives priority to produce that is close to spoiling, and it can suggest nearby buyers, discounts or priority pickup. A transport optimization module groups pickups from nearby farmers and helps build efficient delivery routes.

Blockchain is used only for the events that matter most: harvest, quality certification, transfer of produce, delivery confirmation and proof of payment. This gives a record that is very hard to alter. The expected outcome is a web-based prototype with separate interfaces for farmers, buyers, transporters and administrators. Overall, the platform brings marketplace features, AI decision support, logistics optimization and blockchain traceability together in one supply-chain system.

INTRODUCTION

Agricultural supply chains connect farms to markets, but the flow of crop information, orders, transport and payments is often broken into disconnected pieces. This is a bigger problem for perishable crops, because the time in which they can be sold at a good price is short. A farmer may grow a large quantity with no buyer lined up, while a buyer may need a specific quantity and quality within a fixed time.

AgriFlow AI tackles this coordination problem by giving the main players in the supply chain a single platform. Farmers can publish crops that are expected or already harvested, buyers can post their demand and reserve future produce, and transporters can manage their pickups and deliveries. Administrators can keep an eye on users, orders, payments and analytics. The whole platform is built around a demand-linked workflow. Instead of treating harvesting, selling and transport as separate steps, the system links them through bookings and AI recommendations. This lets farmers plan their harvest around real or predicted demand, and makes it easier for buyers to find suitable supply.

The project also takes the special needs of perishable products seriously. Shelf-life monitoring flags produce that has to be moved quickly. Transport planning looks at where farmers are located and groups their pickups, and blockchain records selected events to make traceability stronger. The larger aim is to move from a scattered marketplace to an integrated decision platform for the supply chain.

LITERATURE REVIEW OF EXISTING SYSTEMS

Several digital platforms support different parts of the agricultural supply chain, including trading, procurement, logistics, advisory, payments and market information. However, each platform has a different primary focus, and their publicly described services do not combine all the functions proposed in AgriFlow AI in a single workflow. This section reviews seven existing systems: e-NAM, AgriBazaar, Ninjacart, DeHaat, Bijak, WayCool and AgroStar.

1. e-NAM (National Agriculture Market)

Overview: A pan-India electronic trading platform connecting APMC markets, with online trading, price discovery, quality-based bidding, market information and online payments.

Limitation: It focuses on market integration and trading. It does not offer AI-based farmer–buyer matching, shelf-life wastage prevention, pickup grouping or blockchain-based event tracking.

Relevance to AgriFlow AI: AgriFlow AI extends trading with demand forecasting, pre-harvest booking, AI matching, perishability management, transport and traceability.

2. AgriBazaar

Overview: A digital marketplace for farmers, traders, processors and exporters, supporting commodity trading, quality assessment, logistics, negotiation and digital payments.

Limitation: Its publicly described services do not include integrated AI farmer ranking, demand forecasting, shelf-life monitoring or blockchain-based event verification.

Relevance to AgriFlow AI: AgriFlow AI adds AI decision support and links demand, perishability, transport and traceability in one workflow.

3. Ninjacart

Overview: A managed supply chain from the pre-harvest to the post-harvest stage, covering procurement, logistics, quality management, payments, demand forecasting and supply planning.

Limitation: It is a managed ecosystem rather than a role-based platform where buyers pre-book harvests and receive AI-ranked farmer recommendations.

Relevance to AgriFlow AI: AgriFlow AI brings similar coordination into one platform, with explicit AI matching, shelf-life management and blockchain traceability.

4. DeHaat

Overview: An end-to-end farmer service ecosystem covering agricultural inputs, soil testing, advisory, financial services, drone spraying and market linkages.

Limitation: Its focus is broad farmer support, not a single workflow that combines pre-booking, AI matching, shelf-life alerts and transporter route planning.

Relevance to AgriFlow AI: AgriFlow AI concentrates on coordinating supply, demand, transport and delivery, with AI support and traceability.

5. Bijak

Overview: A digital B2B trading platform offering market prices, e-negotiation, logistics and payment services.

Limitation: It mainly facilitates trading and transactions. Its publicly described services do not include demand forecasting, AI-based farmer matching, shelf-life monitoring or blockchain traceability.demand forecasting, AI-based farmer matching, shelf-life monitoring, wastage prevention and blockchain-based traceability that AgriFlow AI proposes.

Relevance to AgriFlow AI: AgriFlow AI uses AI to help select suppliers and links transactions with crop freshness, demand, transport and traceability.

6. WayCool

Overview: A managed sourcing, distribution and logistics network for agricultural and food products.

Limitation: It does not publicly describe pre-harvest booking, AI-based farmer–buyer ranking, shelf-life risk alerts or blockchain recording of supply-chain events.

Relevance to AgriFlow AI: AgriFlow AI is a software platform that adds AI matching, forecasting, perishability management and blockchain traceability.

7. AgroStar

Overview: Agricultural advisory, AI-based crop diagnosis, weather insights, inputs, market linkages and farm-to-fork traceability for its produce business.

Limitation: Its main focus is advisory, inputs and market access, not the full workflow from buyer requirements and pre-booking to pickup optimization and blockchain records.

Relevance to AgriFlow AI: AgriFlow AI coordinates the commercial supply chain from crop listing and booking to transport, delivery and traceability.

Summary Comparison:

Platform	Main Focus	Gap Addressed by AgriFlow AI
e-NAM	Online trading across APMC markets	No AI matching, shelf-life alerts or traceability
AgriBazaar	B2B marketplace and logistics	No integrated AI ranking, forecasting or blockchain
Ninjacart	Managed supply chain	No buyer pre-booking with AI-ranked farmers
DeHaat	Farmer services and inputs	No single booking-to-delivery workflow
Bijak	B2B trading and payments	No forecasting, AI matching or traceability
WayCool	Sourcing and distribution	No AI matching, shelf-life alerts or blockchain
AgroStar	Advisory, inputs and market access	No pickup optimization or blockchain event records

Research Gap:

The real gap is not a lack of individual technologies. What is missing is one integrated workflow where demand, supply, perishability, transport, payments and traceability all work together. AgriFlow AI addresses this by proposing a single SaaS architecture in which these functions share information and support one end-to-end agricultural supply-chain process.

Proposed Contribution:

AgriFlow AI offers an integrated platform rather than a single prediction model. Several decision-support components work around one shared supply-chain data model. AI-based farmer–buyer matching ranks the available supply against what the buyer needs. Demand forecasting estimates future crop needs from past orders and seasonal patterns. Shelf-life monitoring flags harvested produce whose risk of spoilage is rising, and wastage prevention suggests actions such as urgent sales, nearby buyers, discount offers and priority pickups. Transport optimization groups nearby pickups and supports efficient delivery planning. Blockchain records key events such as harvest, transfer, delivery and payment verification. Digital orders, payments, invoices and notifications tie all of these activities together on one platform.

In this design, AI, logistics and blockchain work as complementary layers. AI handles prediction and decision support, the logistics modules plan movement, and blockchain provides traceability and verification.

MOTIVATION

Agricultural produce is time-sensitive. A crop loses value when it sits unsold or when transport is delayed. One of the main motivations for this project is to close the gap between what farmers grow and what buyers actually need.

Five practical needs drive the project:

- **Demand-linked harvesting:** farmers should be able to see what buyers need before they harvest.
- **Faster matching:** buyers should find suitable farmers using several factors instead of comparing listings by hand.
- **Reduced wastage:** perishable crops should get attention based on how much shelf life they have left and how urgent they are.
- **Efficient transportation:** pickups should consider farmer locations and delivery needs so there is less unnecessary travel.
- **Transparent traceability:** important events should be recorded in a way that is hard to tamper with, so stakeholders can verify the journey of a crop.

By meeting these needs together, AgriFlow AI aims to turn a fragmented supply chain into a coordinated digital workflow, where every stakeholder can see the information needed for their next decision.

PROBLEM DEFINITION

A. Problem Statement:

Farmers often harvest crops without confirmed buyers, which leads to spoilage, wastage and lower prices. Buyers, on the other hand, may find it hard to get the crop, quantity, quality and delivery time they need. Transport and supply-chain records are also scattered. This project therefore defines the problem as designing and developing an integrated SaaS platform that connects farmers, buyers, transporters and administrators; supports pre-harvest booking; uses AI for matching and forecasting; monitors perishability; optimizes transport; and keeps blockchain-based traceability for important events.

B. Objectives:

The main objectives of the proposed system are:

- To connect farmers, buyers and transporters through one digital platform.
- To allow buyers to pre-book crops before harvesting.
- To help farmers plan their harvest based on actual or predicted demand.
- To reduce wastage through shelf-life monitoring and urgent-sale recommendations.
- To build AI-based farmer–buyer matching that uses price, quality, quantity, distance and delivery requirements.
- To forecast demand from past orders and seasonal patterns.
- To optimize transport and pickup grouping.
- To provide blockchain-based traceability for selected important records.
- To track orders, payments, invoices and delivery status digitally.

C. Proposed Architecture:

Modules & Workflow

1) Farmer Workflow:

Login → Add Crop → Pre-Harvest / Harvested → Receive Booking → Harvest → Pickup → Delivery → Payment

2) Buyer Workflow:

Login → Search Crop → Select Farmer → Book / Buy → Payment → Track Delivery → Confirm

3) Transporter Workflow:

Login → Pickup Request → Optimized Route → Collect Produce → Deliver → Confirmation

4) Admin Workflow:

Login → Monitor Platform → Users → Orders → Payments → AI Analytics → Blockchain Records

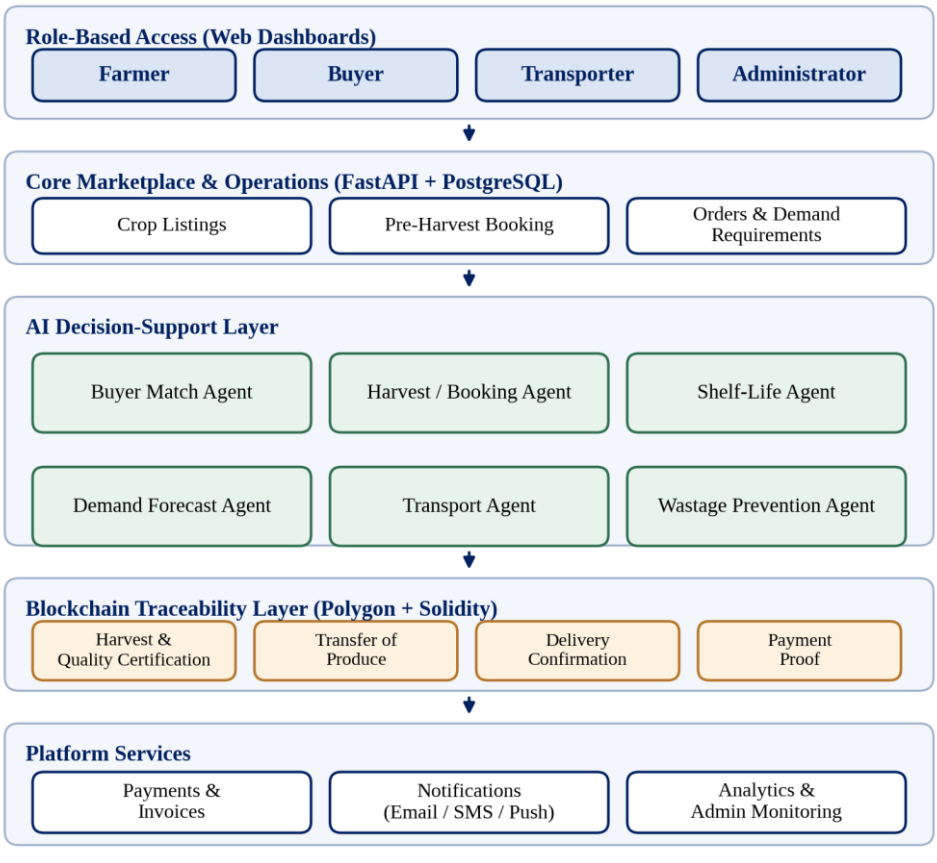


Fig. 1 – Proposed AgriFlow AI System Architecture

The proposed architecture begins with role-based access for farmers, buyers, transporters and administrators. Crop listings, bookings and orders supply the operational data that the AI layer uses. The AI modules handle matching, demand forecasting, shelf-life monitoring, wastage prevention and transport decisions. Blockchain stores selected traceability events, and payments, invoices, notifications and analytics complete the workflow.

AI Modules

1) **Buyer Match Agent:**

   Ranks farmers by price, quality, quantity, distance, delivery time and reliability.

2) **Harvest / Booking Agent:**

   Coordinates pre-harvest booking, scheduled harvest and pickup.

3) **Shelf-Life Agent:**

   Tracks how much usable time a crop has left and gives priority to urgent sales for produce at risk.

4) **Demand Forecast Agent:**

   Predicts future demand from past orders, seasonal patterns and buyer requirements.

5) **Transport Agent:**

   Groups nearby farmers and supports efficient pickup routes.

6) **Wastage Prevention Agent:**

   Spots unsold produce or spoilage risk and recommends immediate action.

D. Experimental Setup:

1. Software Requirements:

- Python for backend logic, AI workflows and data processing.
- Django, Flask or a FastAPI-style REST backend; the proposed implementation uses FastAPI.
- React for the web dashboard and role-based interfaces.
- PostgreSQL for application, order, crop and analytics data.
- PyTorch and other machine-learning libraries to run the AI models.
- Polygon, Solidity and Ethers.js for the blockchain traceability module.
- OpenCV for image or video preprocessing, if future extensions need it.
- Git/GitHub for version control and Docker for reproducible deployment.

2. Hardware Requirements:

- A development workstation or laptop with at least 8 GB RAM; 16 GB or more is recommended.
- A multi-core CPU for application logic and data processing.
- CUDA-capable NVIDIA GPUs can be used for larger AI experiments.
- A stable internet connection for cloud/API deployment and blockchain interaction.

3. Data Requirements:

The core platform needs structured data on farmers, crops, quantities, harvest dates, locations, prices, quality, buyer requirements, orders, payments, transport status and delivery events. Past order data is needed for demand forecasting. In the beginning, the project may use controlled or synthetic records to validate the prototype, and then move to real operational data when it becomes available.

4. Complete Workflow:

The end-to-end workflow of AgriFlow AI is designed as follows:

- A farmer lists a crop with its quantity, expected harvest date, location, expected price and quality.
- A buyer searches for the crop they need or creates a future demand requirement.
- AI ranks suitable farmers using price, quality, quantity, distance, harvest date, delivery requirement and reliability.

- The buyer pre-books a future crop or buys one that is already harvested.
- The platform confirms the expected harvest and pickup schedules, and important supply-chain events are written to the blockchain traceability layer.
- The transport module groups the pickup requests and creates an efficient pickup plan.
- The produce is collected and transported to the buyer.
- The buyer confirms delivery, and the final delivery event is recorded.
- Payment and invoice details are updated digitally.
- The resulting order and demand data can be used to improve future supply and demand planning.

5. Core Data Entities:

Entity	Key Information
Farmer	Profile, location, verification, listed crops, earnings
Crop	Crop type, quantity, quality, price, harvest date, shelf-life status
Buyer	Requirements, orders, booking history, payments
Transport	Pickup requests, route, status, proof of delivery
Order	Buyer, farmer, crop, quantity, price, delivery status
Traceability Event	Event type, timestamp, participants, transaction/hash reference

EVALUATION METHODOLOGY

The system will be evaluated at several levels, so that the individual components and the complete workflow can each be assessed separately.

Module	Evaluation Criteria
Farmer–Buyer Matching	Matching relevance, successful order conversion, ranking quality, and response time.
Demand Forecasting	MAE, RMSE, or MAPE, where suitable data are available.
Shelf-Life / Wastage Prevention	Risk-detection precision, recall, alert timeliness, and reduction in simulated wastage.
Transport Optimisation	Distance travelled, number of trips, pickup consolidation, and estimated transportation cost.
Blockchain	Transaction confirmation time, record integrity, and traceability completeness.
Platform	API response time, end-to-end workflow completion, usability, and resource usage.

Baseline Comparison:

The AI-assisted workflow can be compared with simple rule-based or manual baselines. For example, farmer matching can be compared with picking the nearest farmer, demand forecasting with a historical-average forecast, and transport with ungrouped pickup routing.

Cost Optimisation:

In future, the system could estimate logistics costs using distance, vehicle capacity, number of trips, route length and urgency. The transport module could then compare candidate pickup plans and prefer those that meet delivery constraints at a lower estimated cost.

Sample Integrated Decision:

Example: a buyer asks for 500 kg of tomatoes within a given delivery window. The system can rank farmers by available quantity, quality, price and distance; check whether the crop is already harvested or expected to be ready; estimate the remaining shelf life; group the chosen farmer with nearby pickup requests; and log the key harvest, transfer and delivery events on the blockchain.

Expected Performance Indicators:

- A higher share of orders matched to suitable farmers.
- Lower estimated transport distance and cost through grouped pickups.
- Earlier detection of crops that are close to the end of their shelf life.

TECHNOLOGY STACK

Component	Technology	Purpose
Frontend	React	Role-based dashboards and marketplace interface
UI Styling	Tailwind CSS	Responsive interface styling
Backend	FastAPI / Python	REST APIs and business logic
Database	PostgreSQL	Users, crops, orders, payments and analytics
AI / ML	Python, scikit-learn / PyTorch	Matching, forecasting and decision-support models
Blockchain	Polygon + Solidity + Ethers.js	Traceability records and smart-contract interaction
Maps / Routing	Routing API / optimisation logic	Distance, pickup grouping and route planning
Notifications	Email / SMS / Push service	Booking, pickup, delivery and freshness alerts
Deployment	Docker	Reproducible application deployment
Version Control	Git / GitHub	Source-code management and collaboration

Security & Reliability Considerations:

- Role-based access control will restrict actions according to the farmer, buyer, transporter and administrator roles.
- Sensitive application data will stay in the controlled database instead of being put on-chain unnecessarily.
- Blockchain records will be used to verify selected high-value events.
- Payment and identity workflows should rely on secure authentication and authorized access.
- System logs and audit records can support operational monitoring.

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Project Source:

AgriFlow AI — Smart Agricultural Supply Chain SaaS project specification, supplied for this synopsis. It defines the platform objectives, stakeholder modules, AI agents, blockchain scope, workflow and core features.

APPROVAL STATEMENT

This certifies that the synopsis report titled **"AgriFlow AI – Smart Agriculture Supply Chain SaaS"** has been thoroughly reviewed by the expert panel members of the CSE (Data Science) Department at D. Y. Patil College of Engineering & Technology, Kolhapur. The content has been checked for accuracy, and the proposed work has been approved for further development.

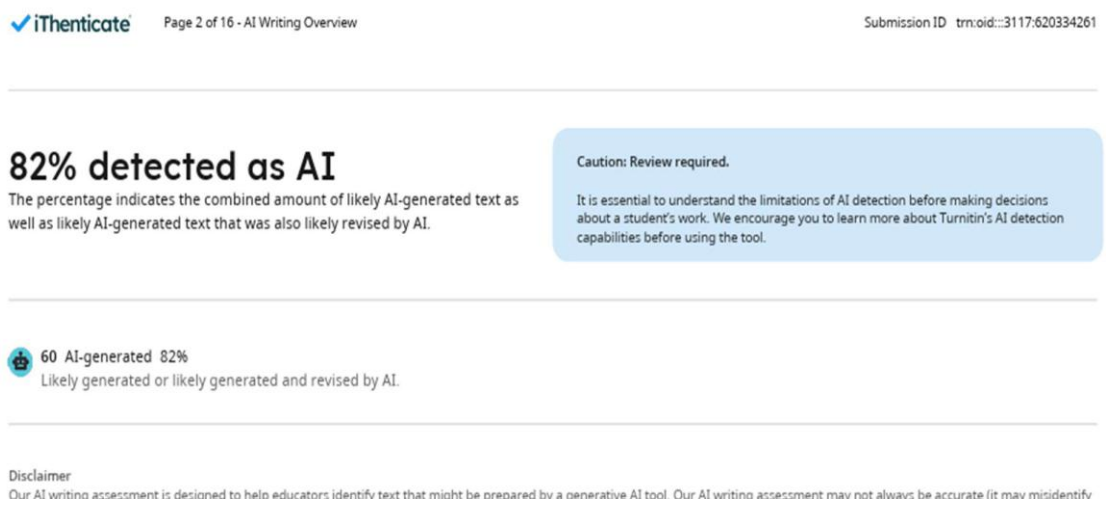
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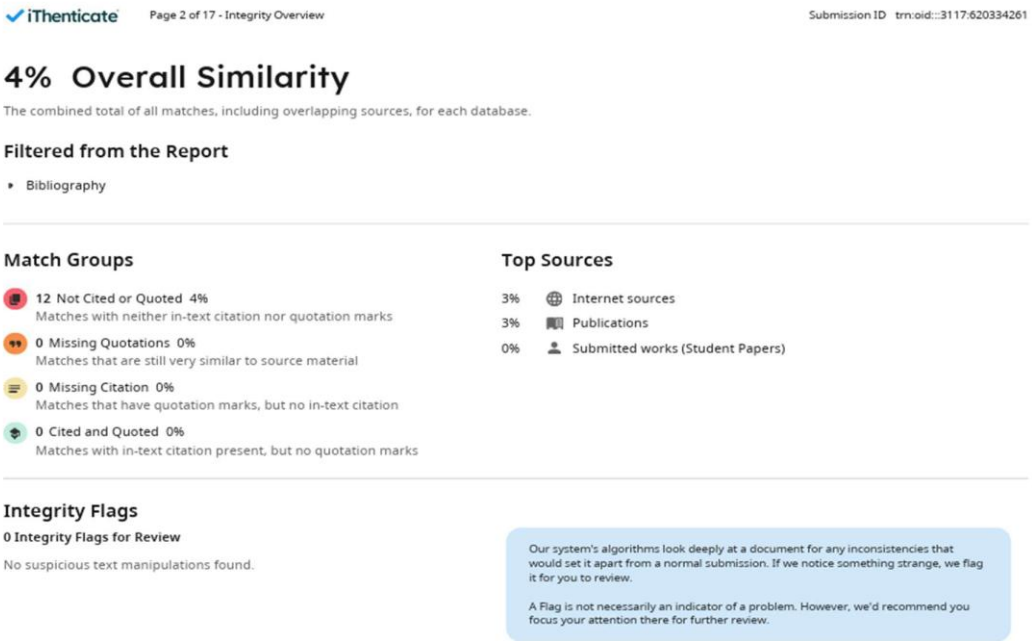
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